

Daniel Hardesty Lewis

AI/ML Engineer – LLM API integration, Python backend, cloud-native AI systems

Specializing in scalable Python backends integrating LLM APIs (OpenAI, Claude, Gemini) with cloud-native AI systems. Delivered production ML across tabular, geospatial, and time-series workloads at supercomputer scale.

Core Expertise

- 8 years **ML Engineering**: Design and operate ML training and inference infrastructure at scale, from data ingestion through model deployment, monitoring, and retraining
- 8 years **Backend & Data**: Build schemas, ETL, and production APIs (Python/FastAPI); manage data-intensive systems with Postgres, Redis, and Elasticsearch
- 8 years **Distributed Systems**: Scale ML workloads across distributed cloud and HPC systems using containerization and orchestration (Docker, Kubernetes, Ray), with explicit cost-performance tradeoffs
- 10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases
- 8 years **Technical Leadership**: Collaborate across engineering, research, and product to ship; mentor engineers through design reviews, code reviews, and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Hugging Face Transformers; LoRA/QLoRA/PEFT
LLM	OpenAI API; Claude API; Gemini API; embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data	SQL; Pandas; Polars; PyArrow; Parquet; Postgres; Supabase; BigQuery; Redis; MongoDB; Elasticsearch
Infrastructure	Linux; Git; Docker; Kubernetes; Modal; Vercel; Terraform; Ray; Dask; Airflow; CI/CD; observability
Additional	CUDA; C++; TypeScript; React; REST/async; GitHub Copilot; ChatGPT API; v0

Experience

Professional

- Nov. 2024 – Present **Co-founder, [Homecastr](#)**, New York, NY
 - Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation.
 - Architected a nationwide neighborhood-level diffusion forecasting model, **matching Zillow's 7.3% 0-year error benchmark at a 1-year forecast horizon** and maintaining a **35.6% median error over 4 years**.
 - Engineered architecture incorporating **spatial inducing tokens, per-horizon loss normalization**, and **DDIM sampling** to balance predictive precision with calibrated price trajectories.
 - Developed predictive validation optimizing PIT uniformity, interval coverage, and sharpness to **rigorously validate forecast reliability** before production delivery.
 - Owned the end-to-end pipeline from data ingestion to real-time serving and refined product strategy through direct feedback from **a16z and MetaProp**.
- Nov. 2024 – Present **Co-founder, [PoliBOM](#)**, Seattle, WA
 - AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation
 - Specified an agentic workflow for BOM parsing and tariff simulation using Airflow, Elasticsearch embeddings, and FastAPI on Supabase and Redis.
 - Directed development of a **conversational AI interface** enabling decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
 - Executing customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.
- Nov. 2023 – Present **Founder, [Summit Geospatial](#)**, New York, NY
 - High-resolution elevation data and web delivery for engineering and hazard workflows.
 - Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
 - Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.

500 RIVERSIDE DR – NEW YORK, NY 10027 – UNITED STATES

☎ +1 (713) 371-7875 • ✉ daniel@homecastr.com • 🌐 dlewis.ai • in [dhardestylewis](#)
🔗 [dhardestylewis](#) • [Google Scholar](#)

- Aug. 2021 – **Senior Data Scientist & Technical Lead**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
- Aug. 2023
- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the \$40M [Texas Disaster Information System \(TDIS\)](#), delivering precision decision-support for civil engineering.
 - Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating [Real-Time Inundation Mapping](#) and TDIS.
 - Scaled climate and flood models on supercomputers, orchestrating million-node distributed jobs across containerized HPC clusters while managing multi-million-dollar compute budgets.
 - Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
 - Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
 - Contributed fundamental research supporting Paola Passalacqua's [2022 EGU Bagnold Medal](#), one of geomorphology's highest honors.
 - Served as Technical Reviewer for [Frontera Doctoral Fellows](#) (2023) and [Large-Scale Computing Pathways](#) (2022) applications, evaluating \$10M+ in computing resource allocations on one of the world's largest supercomputers.
- Feb. 2018 – **Data Scientist & Research Engineer**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
- Jul. 2021
- Competed in [DARPA World Modelers](#) to improve disaster resiliency and food security modeling in East Africa.
 - Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
 - Received research recognition from AAAI President Yolanda Gil during her [farewell address](#) for contributions to automated scientific modeling.

Research

- May 2025 – **Research assistant to Dir. Electrical Engineering Zoran Kostic**, [Columbia University](#), New York, NY
- Aug. 2025
- Math benchmarking and finetuning of Deepseek's GRPO algorithm
- Jan. 2025 – **Research assistant to Dir. Financial Engineering Ali Hirsra (Industry-sponsored research)**, [Columbia University](#), New York, NY
- Oct. 2025
- Developed a multi-asset CVAE latent factor model for **tabular financial data** with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
 - Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.
- July 2019 – **Summer research intern to Pres. AAAI Yolanda Gil**, [University of Southern California](#), Los Angeles, CA
- Aug. 2019
- Integrated hydrological models into the [MINT Platform](#) during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

- Jan. 2018 – **Co-instructor**, [The University of Texas at Austin](#), Austin, TX
- Dec. 2020
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

Select Publications

- 2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM Tüs*, 2021. DOI: [10.1145/3453172](#)
- 2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: [10.1145/3308557.3308711](#)

Education

- expected 2026 **M.S., Urban Planning**, [Columbia University](#), New York City, NY
- 2017 **B.S., Pure Mathematics**, [University of Texas](#), Austin, TX

Key Projects

- oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

Activities

- Sailing Volunteer Atlantic SAR of the [Cheeki Rafiki](#) Bikepacking US West & East Coasts

500 RIVERSIDE DR – NEW YORK, NY 10027 – UNITED STATES

☎ +1 (713) 371-7875 • ✉ daniel@homecastr.com • 🌐 [dlewis.ai](#) • in [dhardestylewis](#)
 🔗 [dhardestylewis](#) • [Google Scholar](#)

Languages

Spanish Native Proficiency

French Professional Proficiency

500 RIVERSIDE DR – NEW YORK, NY 10027 – UNITED STATES

📞 +1 (713) 371-7875 • ✉ daniel@homecastr.com • 🌐 dlewis.ai • **in** [dhardestylewis](#)
👤 [dhardestylewis](#) • [Google Scholar](#)